

BIOLOGY FORM FOUR – MARKING SCHEME

SECTION A (16 Marks) – 1 mark each

1. A – Hypogeal germination
2. A – They reproduce sexually by seeds
3. B – Medulla oblongata
4. A – Cut the wound
5. B – They are only active inside living cells
6. D – B+ or B–
7. B – Pupil becomes large
8. B – Speaking respectfully
9. A – Auxin, cytokinin, gibberellin and abscisic acid
10. D – Test cross

SECTION B (54 Marks)

Question 3

(a) Qualities of a good laboratory (Any four) – 4 marks

- Good ventilation
- Adequate lighting
- Enough space
- Water supply
- Strong benches
- Storage facilities
- Safety equipment

(b)

(i) First Aid – 1 mark

(ii) Actions (Any four) – 4 marks

- Calm the victim
- Immobilize the limb
- Remove jewellery
- Keep victim still

- Wash wound
- Take victim to hospital

Question 4

(a) Tuberculosis – 1 mark

(b) Mycobacterium tuberculosis – 1 mark

(c) Prevention (Any four) – 4 marks

- BCG vaccination
- Good ventilation
- Avoid overcrowding
- Cover mouth when coughing
- Early treatment
- Proper nutrition

(d) Advice (Any two) – 2 marks

- Seek hospital treatment
- Take medicine regularly
- Eat balanced diet
- Avoid spreading infection

Question 5

(i) Reflex action – 1 mark

(ii) Reflex arc – 1 mark

(iii) Diagram components – 5 marks: receptor, sensory neuron, spinal cord/relay neuron, motor neuron, effector

(iv) Significance – 2 marks: protection from injury, quick response

Question 6

(a) Earthworms as farmers' friends (Any three) – 3 marks

- Improve aeration
- Improve fertility
- Mix soil
- Decompose organic matter

(b) Hermaphrodite: has both male and female organs – 2 marks

(c) Arthropod success features (Any four) – 4 marks

- Exoskeleton
- Jointed appendages
- Adaptability
- Efficient respiration
- High reproduction

Question 7

(a) Reasons

- Moist respiratory surface – allows gas diffusion (2 marks)
- Ileum long and folded – increases absorption surface (2 marks)
- Rough laboratory floors – prevent slipping (2 marks)

(b) Skin functions (Any three) – 3 marks: protection, temperature regulation, excretion, sensation

Question 8

(a) Muscle cramp – 2 marks

(b) Causes (Any two) – 2 marks: lactic acid, overexertion, dehydration, lack of oxygen

(c) Prevention (Any two) – 2 marks: warm-up exercises, hydration, balanced diet, rest

SECTION C (30 Marks)

Answer any TWO questions – each question 15 marks.

Question 9 – Birth control methods

- Prevent union of gametes: condom, diaphragm, cervical cap, spermicides
- Suppress gamete formation: contraceptive pills, hormonal injections
- Prevent implantation: IUD, emergency pills, implants

Question 10 – Risky behaviours among youths

Expected points: drug abuse, alcohol, peer pressure, early sex, poverty, media influence.

Consequences: HIV/AIDS, pregnancy, dropout, crime, health problems.

Question 11 – Factors making a person look older

Examples: poor nutrition, stress, smoking, alcohol, lack of sleep, disease, sun exposure, hard working conditions.